

Week 1 Exercises

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Please complete all exercises below WITHOUT using any libraries/packages.

Exercise 1

Assign 10 to the variable x. Assign 5 to the variable y. Assign 20 to the variable z.

```
#Assign values to variables
x <- 10
y <- 5
z <- 20
```

Exercise 2

Show that x is less than z but greater than y.

Note: your output must be a SINGLE boolean, do not output a boolean for each expression.

```
#Single boolean expression to check if x is less than z and greater than y
result_ex2 <- x < z & x > y
result_ex2
```

```
## [1] TRUE
```

Exercise 3

Show that x and y do not equal z.

Note: your output must be a SINGLE boolean, do not output a boolean for each expression.

```
#Single boolean expression to check if x and y do not equal z
result_ex3 <- x != z & y != z
result_ex3
```

```
## [1] TRUE
```

Exercise 4

Show that the formula $x + 2y = z$.

Note: your output must be a **SINGLE** boolean

```
#Check if x + 2y equals z in a single boolean expression
result_ex4 <- (x + 2 * y) == z
result_ex4
```

```
## [1] TRUE
```

Exercise 5

I have created a vector (test_vector) of integers for you. Determine if any of x, y, or z are in the vector.

Note: your output must be a **SINGLE** boolean, do not output a boolean for each expression.

```
#Check if x, y, or z are in the test_vector
test_vector <- c(1, 5, 11:22)
result_ex5 <- x %in% test_vector | y %in% test_vector | z %in% test_vector
result_ex5
```

```
## [1] TRUE
```

Exercise 6

Show which value is contained in the test vector. To do this you will need to create an element-wise logical vector using operators. $x == \text{vector}$. Once you have done that you will need to use slicing to return all indices that have matches. **Note:** your output should be two integers

```
#Find which values (x, y, or z) are contained in the test_vector and their indices
matches_x <- which(test_vector == x)
matches_y <- which(test_vector == y)
matches_z <- which(test_vector == z)

#Combine all matching indices into a single vector
all_matches <- c(matches_x, matches_y, matches_z)
all_matches
```

```
## [1] 2 12
```